

Can we predict oil prices?

(With Deep Learning)

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Background: Oil price fluctuation

Supply Factors

- Production decisions
- Geopolitical conflicts and wars
- Pipeline or shipping disruptions

Demand Factors

- Global economic growth
- Industrial activity
- Seasonal energy consumption

External shocks

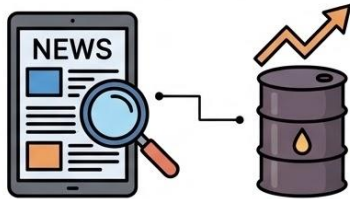
- Extreme weather
- Financial speculation in energy markets
- Government policies and regulations

Predicting oil prices is almost impossible in real energy markets due to multiple variables interacting simultaneously, some hidden, some with delayed effects

=> machine learning(especially deep learning) models can be useful for analyzing many variables at the same time and identifying patterns that influence oil price changes

Project Overview

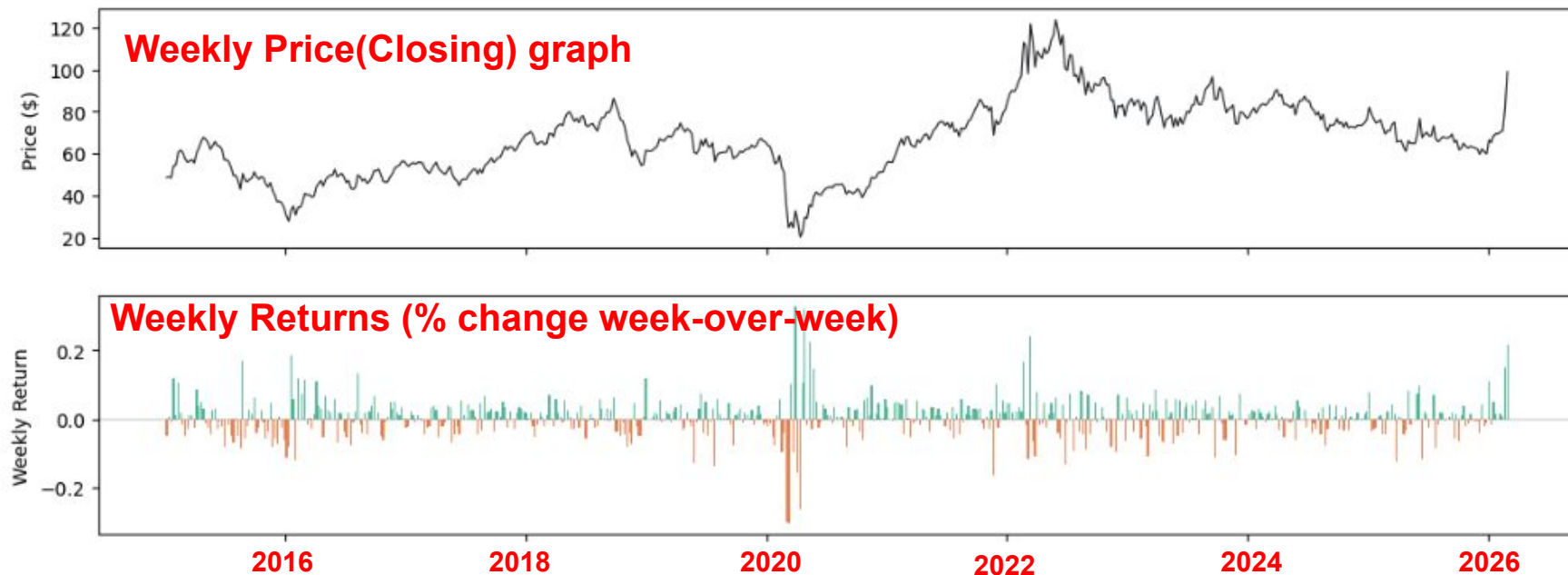
- **Deep Learning** is one category of **Artificial Intelligence** that utilizes a special network resembling a human brain. This design enables them to process more complex, vast amounts of data.
- If you create a Deep Learning Algorithm and feed good data into it, the algorithm is able to find relevant **patterns** in the data as instructed (**if there is one**)
- In this project, we tried to find the patterns between **signals in news articles** and **oil prices(Brent Crude)** using a deep learning algorithm called **Transformer**.



The Technical Part..

Oil Price Data: Brent Crude Prices (Yahoo Finance)

Brent: 583 weeks, 2015-01-05 to 2026-03-02



- We used the price data of **Brent Crude Oil** via the Yahoo Finance platform
- Price data from 2015-01-01 ~ 2026-03-02 (**583 weeks**)
- This is the **target variable** we're trying to predict

News Data: GDELT Global Content Analysis Measures (GCAM)



- **GDELT**(Global Database of Events, Language, and Tone) is a massive open-source platform that processes billions of global news articles in real-time and provides data about it.
- From this platform we will get **news data** which we will **use** to predict the target variable(oil prices)

News Data: GDELT Global Content Analysis Measures (GCAM)

- We used GDELT's **GCAM dataset**
- It **quantifies** the underlying **themes and emotions** in each news article, and categorizes them into about 3,000 emotion/theme **dimensions**.
Each dimension is a score of a specific aspect of news tone, such as 'surprise', 'confusion', 'hostility', etc.
- For example, a news article about the current war in Iran could be scored high in dimensions like 'hostility', 'conflict', and low in dimensions like 'peaceful'. The more themes an article covers, the more dimensions it is scored on.

Training

- So what we're trying to do is seeing if the model can **predict** oil prices using **those 3,000 kinds emotional tones(dimensions) in news** articles(Ex. If news articles tends to contain positiveness, will oil prices go up?)
- We collected those dimension data from 1 billion news articles(1,056,029,354) from 2015-01-01 ~ 2026-03-02
- We trained our model to predict the next week's oil price based on emotional tones(or dimensions) from the **past 8 weeks**.
- We computed performance scores to evaluate our model.(We ran the model 100 times and averaged these scores for accuracy)

Results

We couldn't even slightly predict oil prices (At least with our algorithm)

Some (miserable) metrics results ..

- **$R^2 = 0.0015$**
A score that ranges from 0 to 1 // 0 means no predicting ability // 1 means perfect fit.
- **Corr = 0.1777**
Very low correlation between predicted and actual returns
- **Direction Accuracy = 52.7%**
Shows whether the model could at least predict the prices will go up or down;
disregards magnitude // The model correctly predicted whether oil prices would go up or down 52.7% of the time - No different from a coin flip

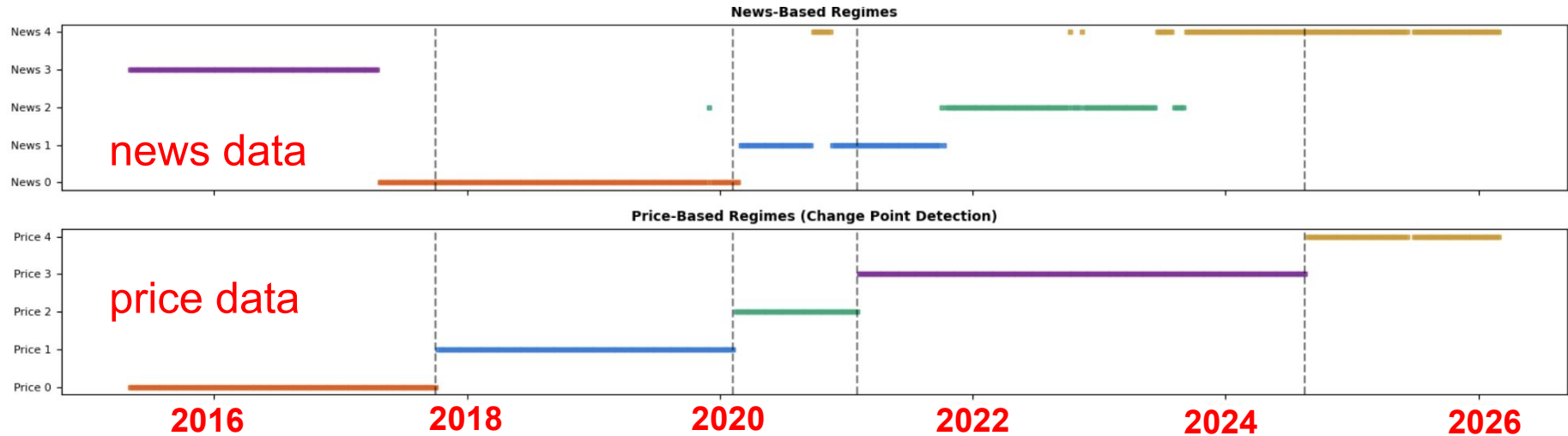
Possible Explanations...

- **The approach itself was wrong**
Focusing on concrete information like supply and demand figures rather than pure emotional signals from news articles could have produced better results. Relying solely on sentiment + a bit of finance metrics was a bit too ambitious
- **1 billion news articles weren't enough**
around 1 million news articles are published every day.. that's only a 1000 days
- **Our model's quality wasn't good enough**
- **It's just impossible to model all the cause-effect relationships that influence oil price changes (Probably the most likely)**

However, there were some other findings →

1. The model was able to identify big patterns in prices

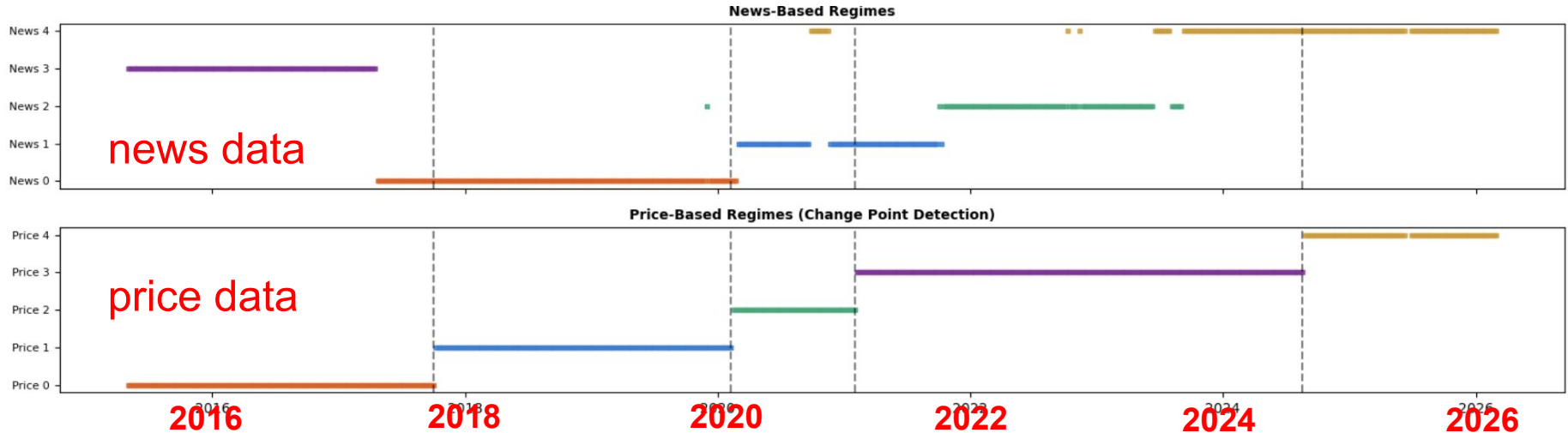
- Above: When we applied a simple grouping algorithm to our model's processed news data, it grouped the news data like below on a timeframe.
- Same color means similar news attributes



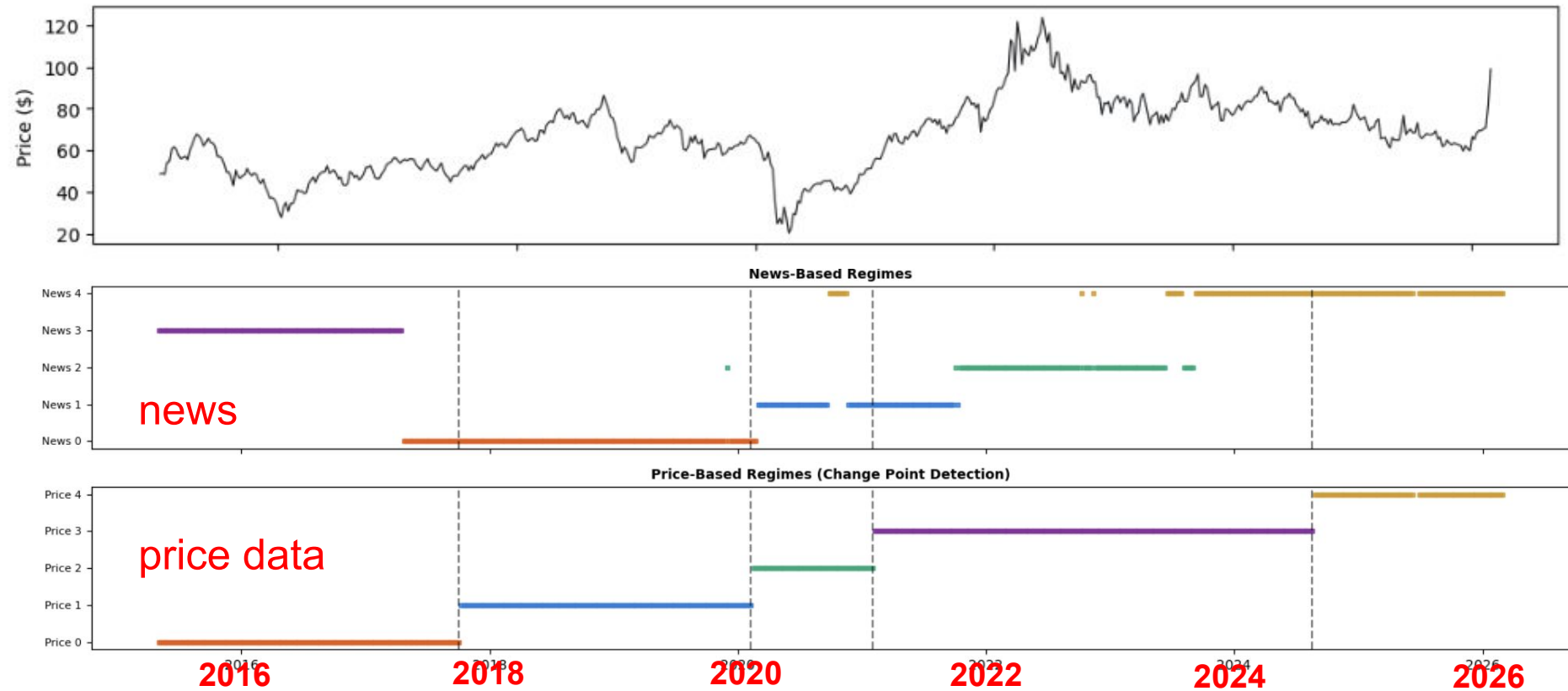
- Below: When we applied a simple grouping algorithm to the Brent crude price data based on price pattern changes, it grouped the price data like above on a timeframe // Same color means similar price patterns

1. The model was able to identify big patterns in prices

- The big patterns in news data lines up with big patterns in oil prices.
- This means from news data alone, the model could tell what kind of market it is, just from how the world talks about it(Even though the model can't predict tomorrow's oil price) // Model was able to provide macroeconomic insights based on solely news data

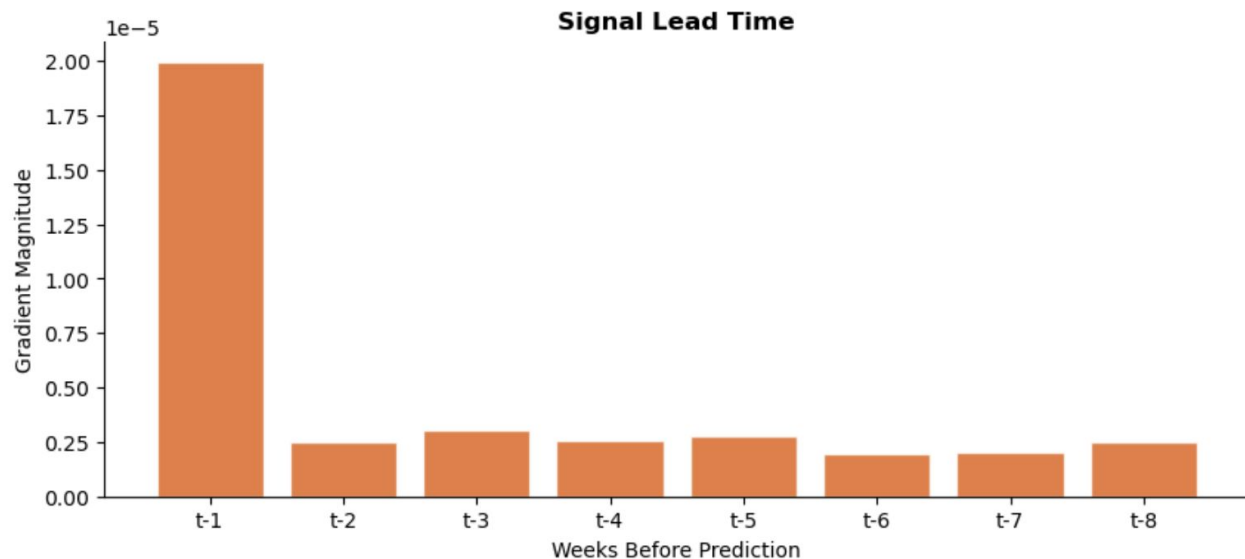


Brent: 583 weeks, 2015-01-05 to 2026-03-02



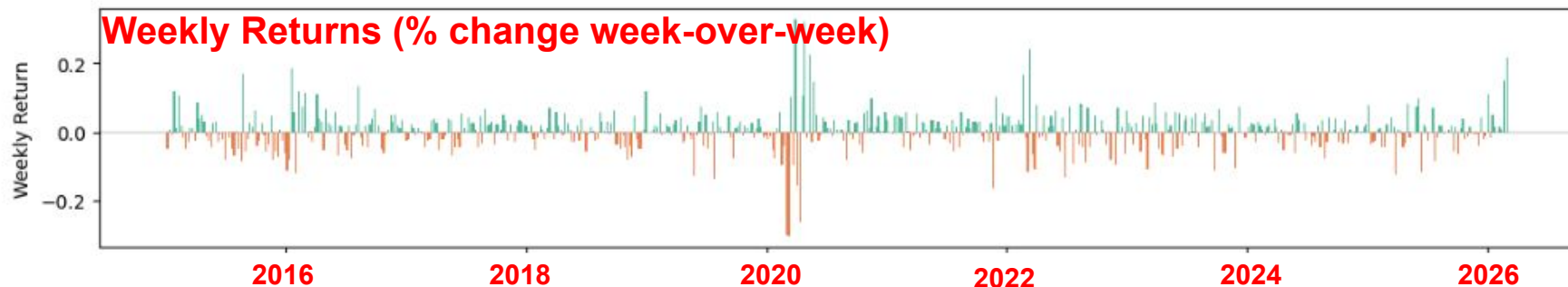
This could happen because news and prices could both reflect the same geopolitical incidents such as war or diseases

2. How quickly does news relate to oil price movements?



- We measured how much each of the 8 input weeks influenced the model's prediction by computing how sensitive the output is to changes in each week's data.
- Result: The most recent week ($t-1$) was $\sim 10x$ more important than all other weeks. Last week's news matter most, and this could also mean news and oil prices react simultaneously to events, rather than news leading prices.

3. Which news dimensions reacted most during price swings?



- Price swings are when prices change most extremely over the week; We defined them as the top and bottom 10% in the weekly return numbers.
- We then measured which of the 3,000 news dimensions spiked or dropped the most during those weeks compared to normal weeks (using the z-score - how many standard deviations it deviated from its normal week average)

During Crashes

Rank	Dimension	Label	Z	Description
1	c9.365	UNCTUOUSNESS	0.991	Oil-related terms/themes
2	c18.172	ENV_OIL	0.939	Oil-related terms/themes
3	c9.265	SMOOTHNESS	0.884	Oil-related terms/themes
4	c9.77	DISPERSION	0.857	Oil-related terms/themes
5	c23.22	Level3Negative	-0.832	Negative sentiment. Decrease - news may shift from emotional language to factual/urgent reporting mode
6	c23.24	Level4Negative	-0.826	Negative sentiment
7	c23.20	Level2Negative	-0.822	Negative sentiment
8	c18.290	ECON_IDENTITYTHEFT	0.821	GDELT "economic crime/identity theft" theme
9	c9.366	OIL	0.814	Oil-related terms/themes
10	c23.18	Level1Negative	-0.808	Negative sentiment
11	v22.7	Valence (Men)	-0.806	How pleasant the news language feels, scored by men
12	c9.853	ECONOMY	0.787	More economic language during crashes is intuitive
13	c9.342	LUBRICATION	0.786	Oil-related terms/themes
14	v22.1	Valence (All)	-0.777	How pleasant the news language feels, scored by all genders
15	c2.10	AUD	0.775	"auditory" words(unclear)

During Rallies

Rank	Dimension	Label	Z	Description
1	c18.295	SOC_POLARIZED	1.287	GDELT "social polarization" theme. Social division discourse increases during oil booms? economic booms may amplify inequality debates
2	c9.365	UNCTUOUSNESS	1.22	Oil-related terms/themes
3	c18.172	ENV_OIL	1.173	Oil-related terms/themes
4	c9.366	OIL	1.116	Oil-related terms/themes
5	c9.342	LUBRICATION	0.947	Oil-related terms/themes
6	c9.271	CLOSURE	0.862	words about physical closing (unclear)
7	c2.42	Decr	0.86	words related to "decrease" maybe things like "decline in unemployment" ? (unclear)
8	v22.7	Valence (Men)	-0.834	How pleasant the news language feels, scored by men Drops in both - even oil surges come with emotional unease
9	v22.1	Valence (All)	-0.803	How pleasant the news language feels, scored by all genders
10	v22.4	Valence (Women)	-0.793	How pleasant the news language feels, scored by women
11	c15.34	belonging	0.79	"belonging" emotion Social bonding language increases during booms?
12	c9.265	SMOOTHNESS	0.785	Oil-related terms/themes
13	c9.853	ECONOMY	0.779	
14	v22.6	Dominance (Women)	-0.778	How much the news language conveys a sense of control, scored by women Drops during rallies — sense of control decreases during price surges.
15	c15.143	helplessness	0.773	oil price surges could create a sense of powerlessness for consumers and importing nations

Summary

1. Deep learning(at least our algorithm) could not predict weekly oil prices, but news data alone could be used to identify macroeconomic market schemes.
2. If news affects oil prices, it happens within a week, and anything older than that adds almost no information.
3. Certain news dimensions react significantly during extreme price swings.

During crashes:

- Negative sentiment decreases in news articles(one explanation: news may shift from emotional language to fact-focused reporting mode)
- 'Emotional pleasantness' of news language drops across all gender groups

During rallies:

- Social polarization discourse increases(one explanation: economic booms may amplify inequality debates)
- 'Emotional pleasantness' drops here too — even surges come with unease
- 'Sense of control' decreases and 'Helplessness' increases

Limitations

- Model was trained on about 400 weeks, which is small by deep learning standards. If we analyzed a larger dataset performance could have improved.
- We could have tried to use a better/complex deep learning structure, and the biggest limitations could be inherent in our algorithm
- Overall, we analyze all data based on weeks, but weekly frequency may be too coarse, since daily data could reveal sharper patterns.

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